

Interactive Explainable Dashboard for Cardiovascular Risk Prediction

This dashboard predicts cardiovascular disease risk and explains the result for patients, clinicians, and machine learning researchers.

Patient Information

Age

20406080100

80

Sex

Male

×

▼

Systolic Blood Pressure

−

140

+

Diastolic Blood Pressure

−

100

+

Cholesterol

Above normal

×

▼

Glucose

Normal

×

▼

Smoking Status

No

×

▼

Alcohol Intake

Yes

×

▼

Physical Activity

Yes

×

▼

BMI

15253550

35

Prediction Result

Estimated Risk Level: High

Predicted Probability: 82.8%

Prediction class: Cardiovascular disease

Patient Explanation	Clinician Explanation	ML Researcher View	Ethics and Safety
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Machine Learning Researcher View

Model used: Gradient Boosting Classifier

Feature importance values are extracted from the fitted Gradient Boosting model.

For deeper explainability, this section can be extended with SHAP summary plots, waterfall plots, and local feature contribution values.

Global Feature Importance for Gradient Boosting

